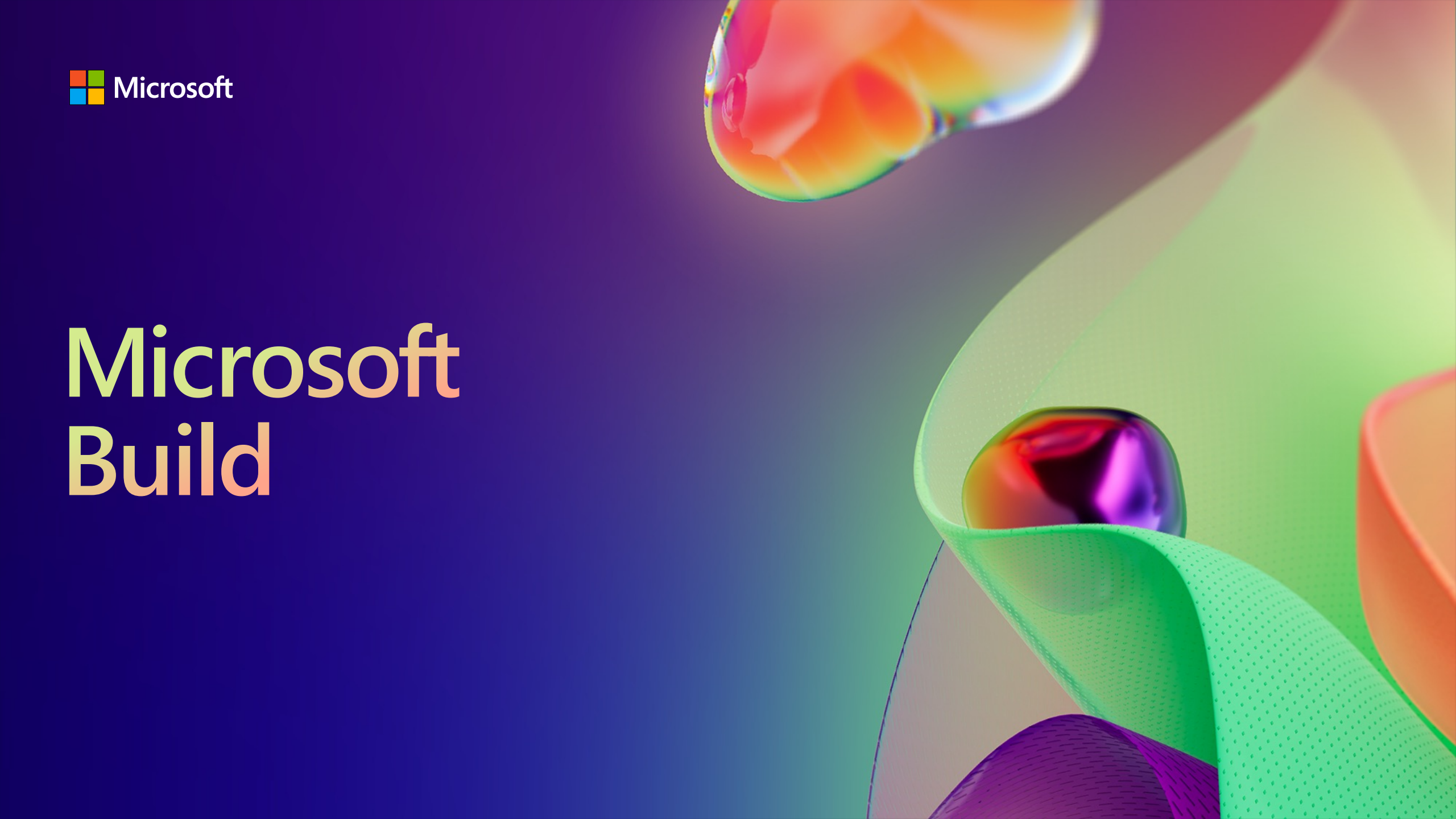

Microsoft
Microsoft
Build

Microsoft

All things client and mobile application development with .NET

David Ortinau | Product Manager | david.ortinau@microsoft.com
Maddy Montaquila | Product Manager | maddy@microsoft.com



2023 .NET MAUI Roadmap

David Ortinau

@davidortinau

david.ortinau@microsoft.com



MODERN CLIENT

David Ortinau

Principal Product Manager

St. Louis, Missouri USA

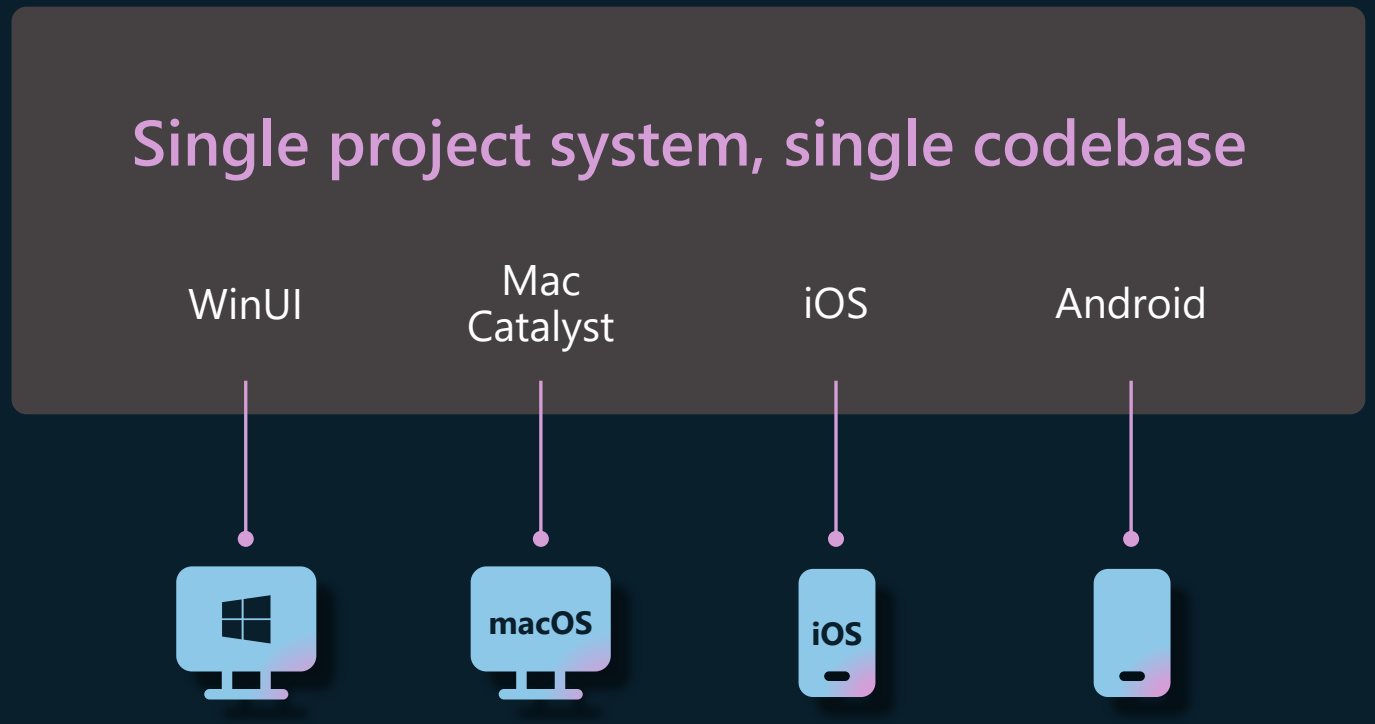
Help customers succeed delivering client app experiences across various screens using .NET SDKs for mobile and desktop cross-platform – aka .NET MAUI (and Xamarin).

I like to trail run and have completed multiple distances up to 50 miles.

Grandfather. Doesn't age.

.NET Multi-platform App UI (MAUI)

Generally available



Unified .NET

Mobile, desktop, hybrid

Productive development

Native APIs, Native performance

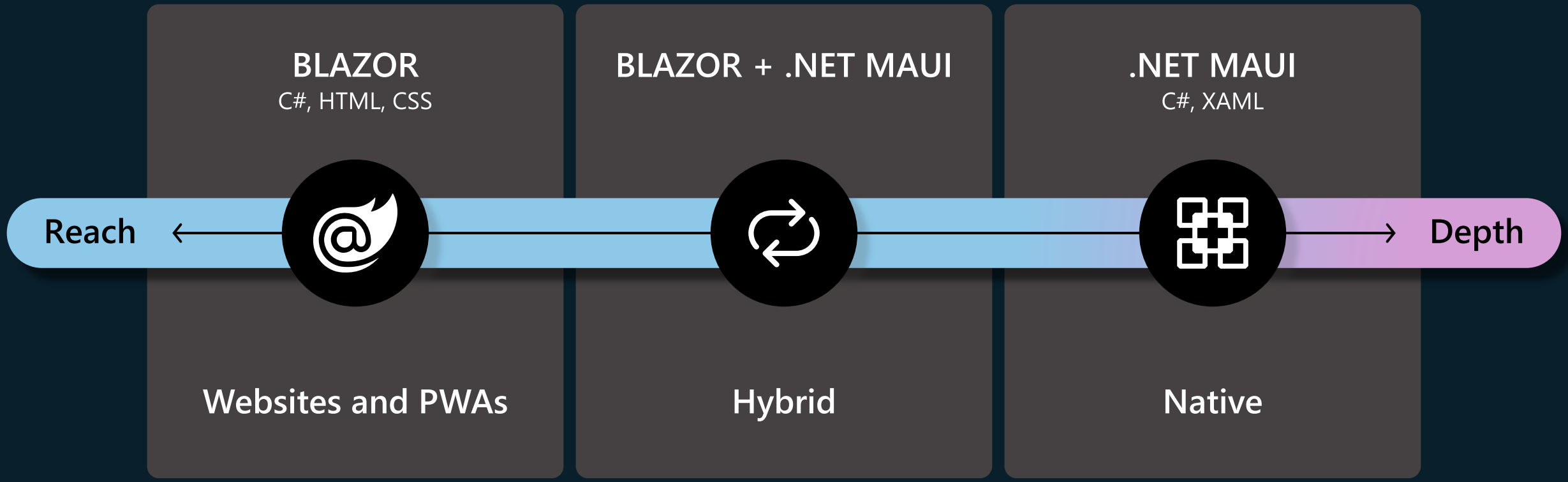
40+ UI controls

Common platform services

Accessible by default

Visit dot.net/maui for more

.NET offers choice for developers









Irth

Actionable insights from the industry's most experienced and trusted.


9:53
95%
Irth


Retail Insight

3M

hogeschool
vives

irth
UtiliSphere

S


Xari

OE
XA

MAHIZ
grow your business

H3Y

ups

CARDIFF
CAERDYDD

Fidelity

Microsoft
Dynamics 365

Microsoft
Azure

CIVICA

SPORTS
NEXT
SPORTS×TECH

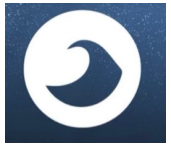
DevInMotion

Belgian
Red Cross
Flanders

ambientindustries

A

insight

esri


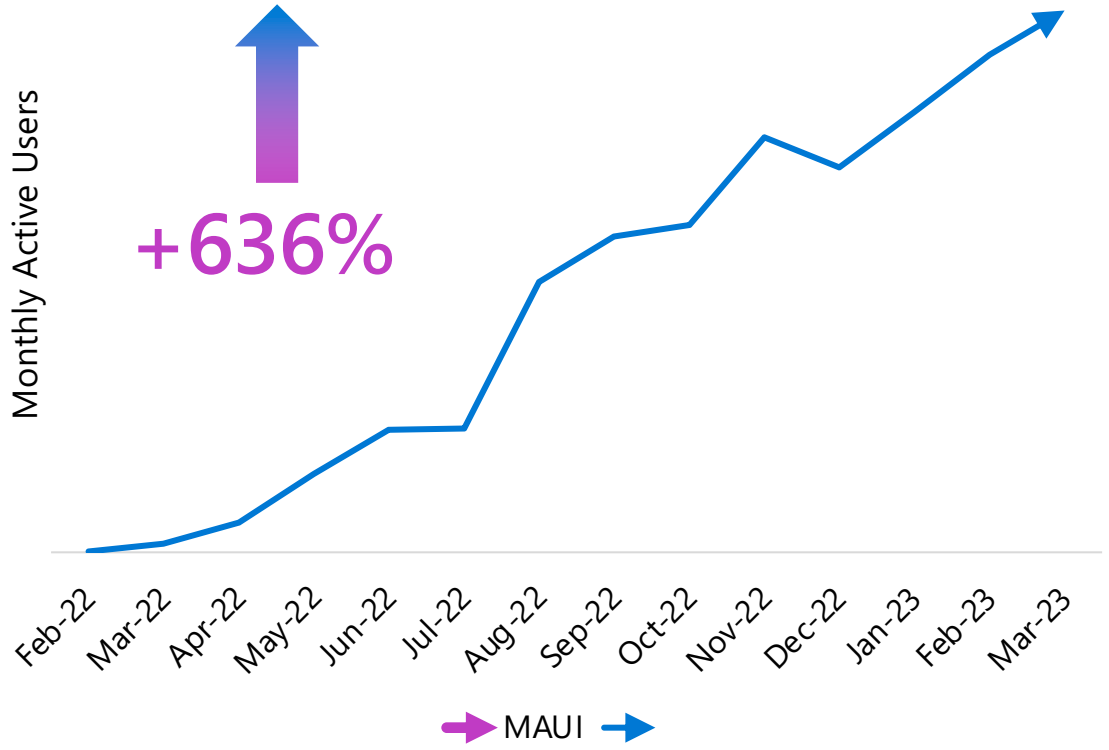
Soluzione 1

REY
TECHNOLOGY

[A]


.NET MAUI is Growing and Active!

.NET MAUI is Growing!



Most active .NET Foundation project

Project	Pull Requests
.NET MAUI	1779
Azure SDK	1460
ASP.NET Core	641
Windows Forms	322
EF Core	212
WPF	199
MSBuild	196
WCF	86

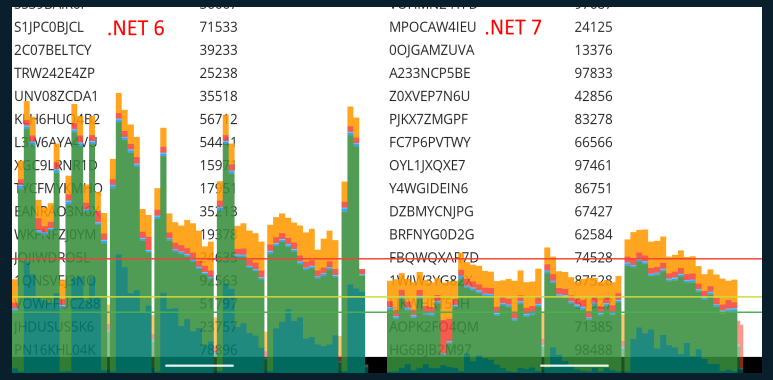
Project	Issues
.NET MAUI	1301
ASP.NET Core	1057
Azure SDK	677
EF Core	468
Windows Forms	181
MSBuild	165
Orleans	86


Thank you contributors!

.NET 7 Performance

^51%

Faster UI Rendering



Smoother Scrolling

545ms

Android Startup

v5%

Smaller iOS App Size

https://aka.ms/maui-7-perf

Visual Studio Improvements

^35%

Pair to Mac Success Rate

1/3 less

Hot Reload Errors

^14%

First Build Success

^12%

First Launch Success

.NET MAUI News Highlights

Supported 1st Party Services

- App Center
  - Crash reporting
  - Monitoring
  - UI Test
- InTune
- MSAL.NET / Azure AD

Productivity

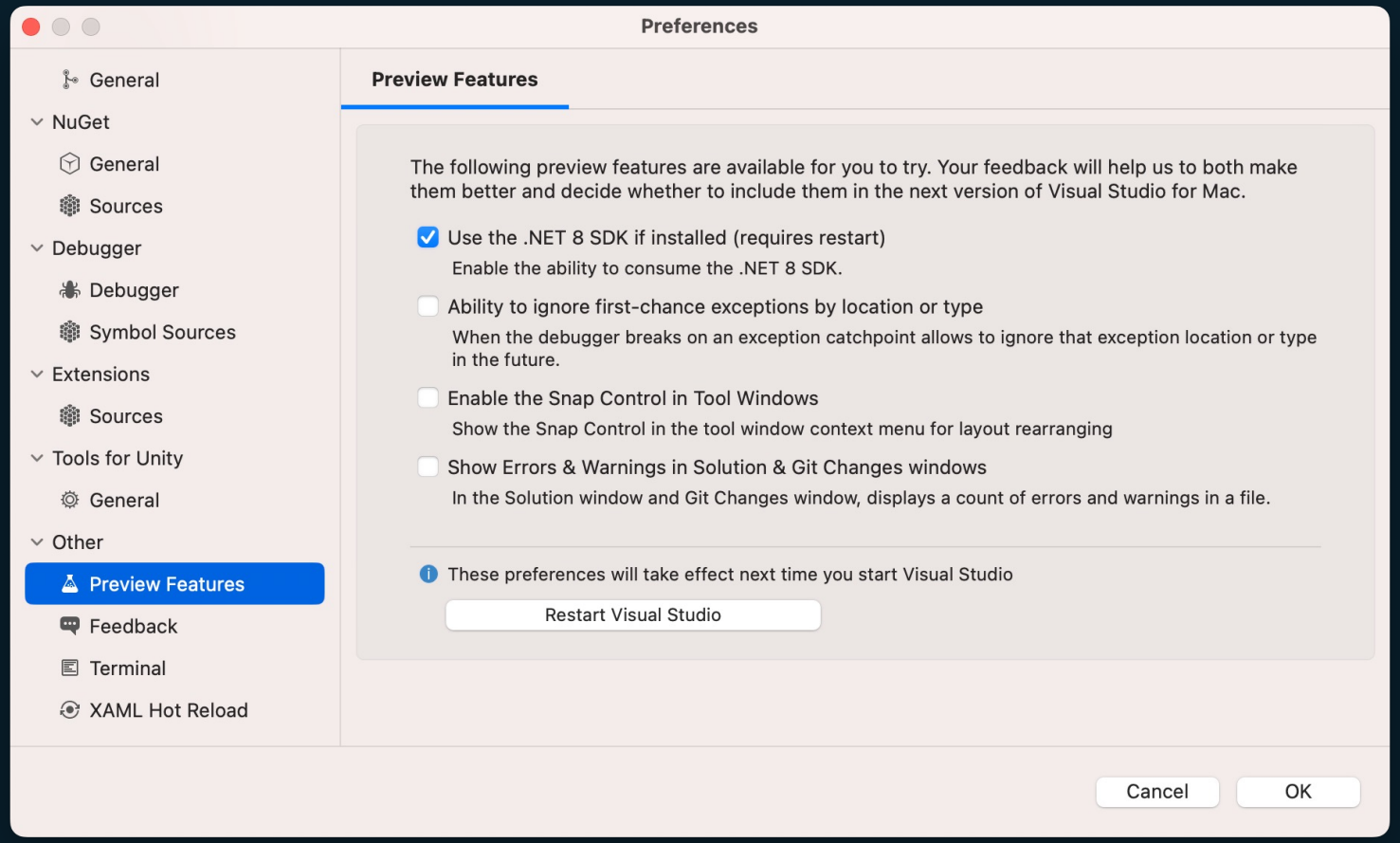
- .NET Hot Reload
- XAML Hot Reload
- Live Preview
- Live Visual Tree
- Hot Restart
- Mac Build host
- iOS Remote Simulator
- VS Code (coming soon)

(Preview)

.NET MAUI in Visual Studio Mac

Highlights:

- Solution explorer
- New project creation
- Debug Android, iOS, macOS
- Intellisense and IntelliCode
- XAML Hot Reload

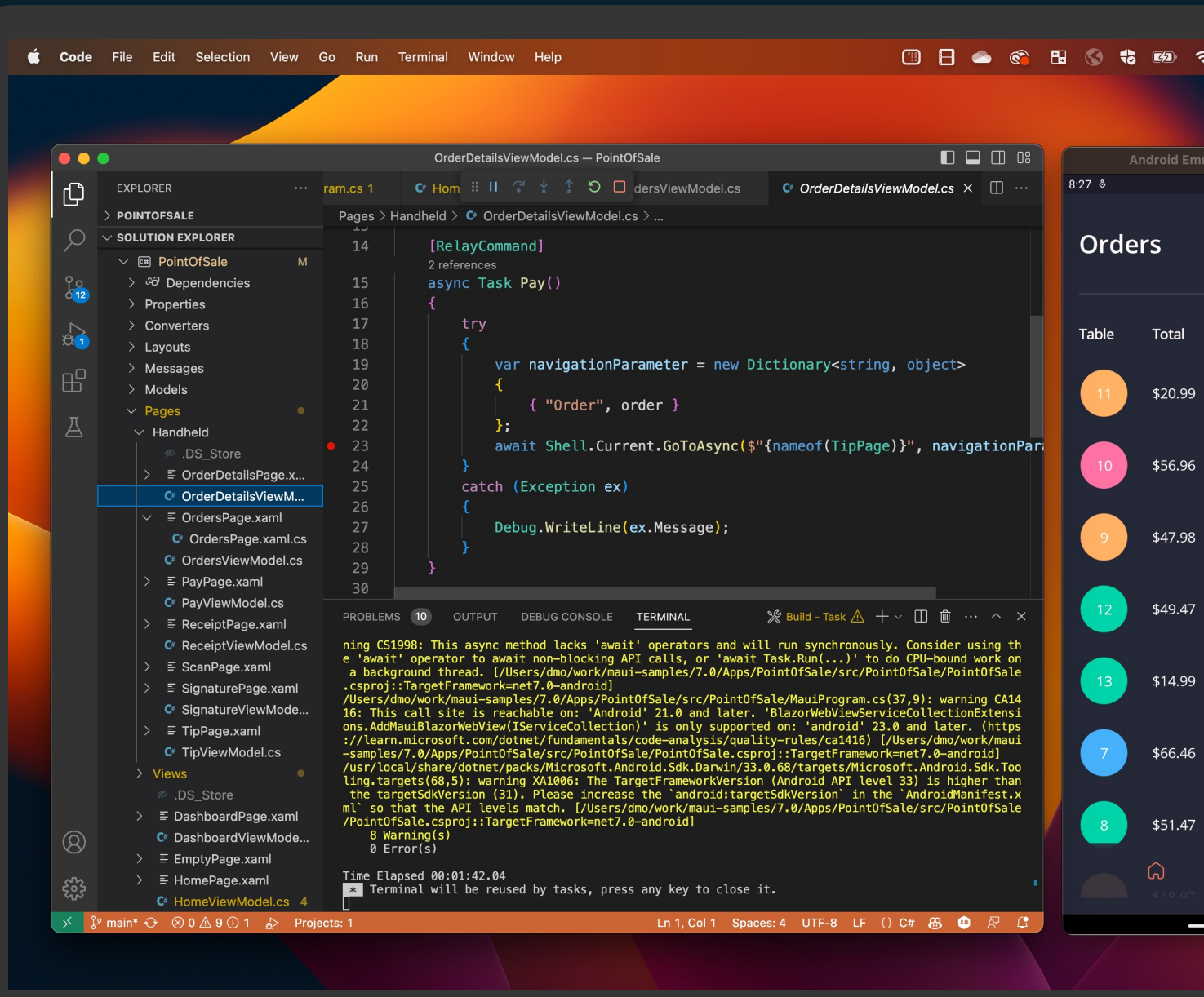


(Preview)

.NET MAUI for Visual Studio Code

Highlights:

- Solution explorer
- New project creation
- Environment configuration
- Debug Android, iOS, macOS, Windows
- Intellisense and IntelliCode

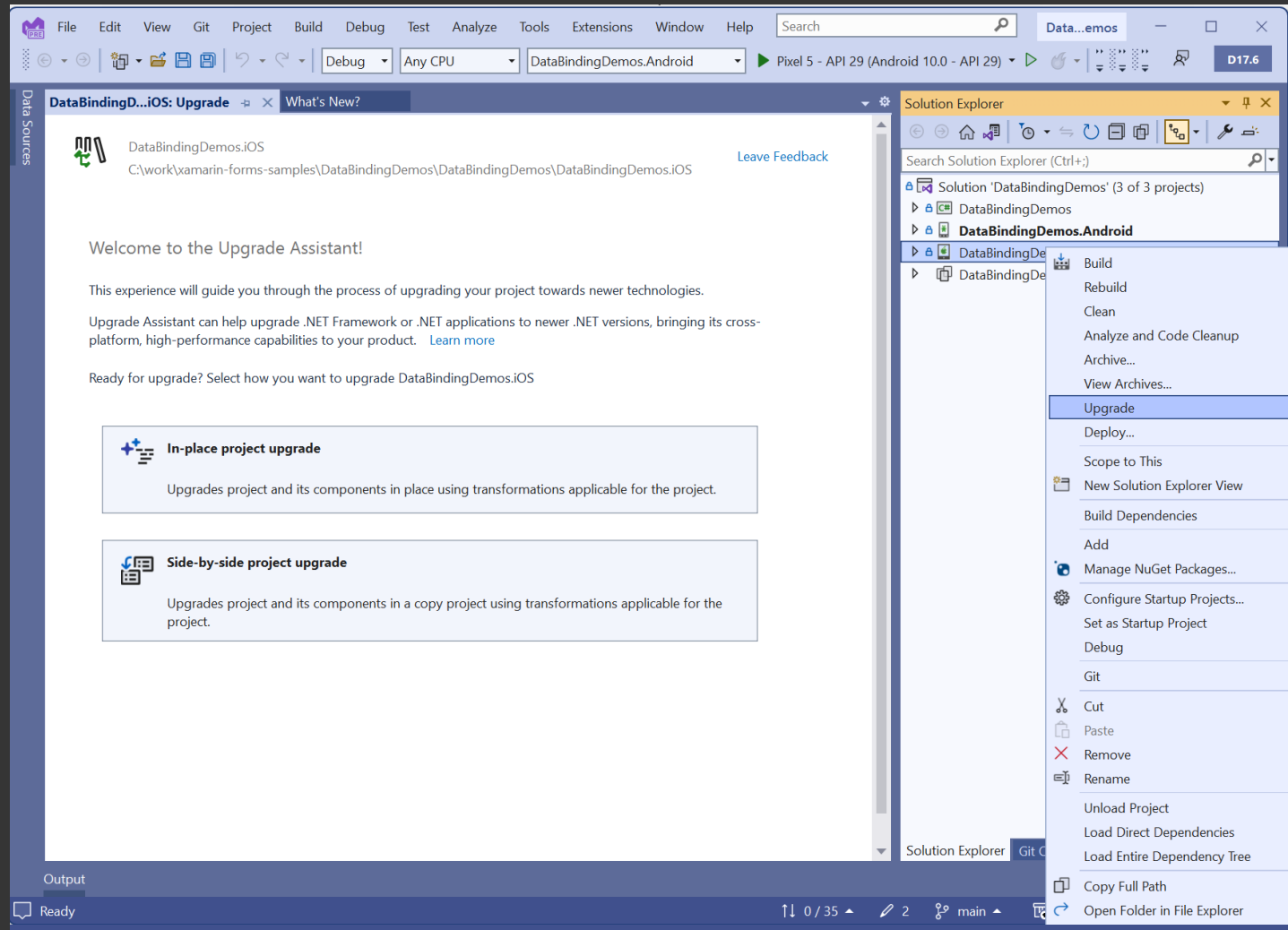


.NET Upgrade Assistant for Xamarin Projects

Highlights:

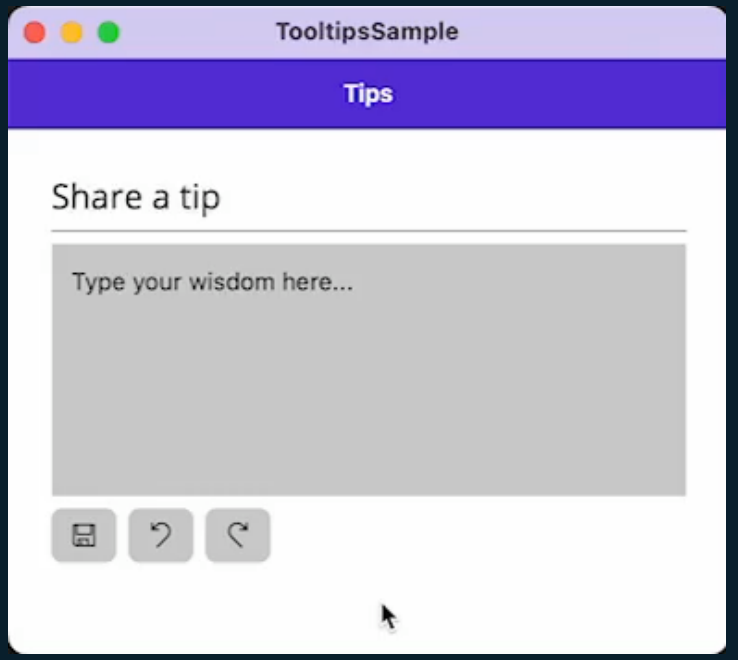
- Converts projects
- Replaces known NuGets
- Fixes namespaces
- Fixes base types
- Applies various code updates

maui-upgrades@microsoft.com



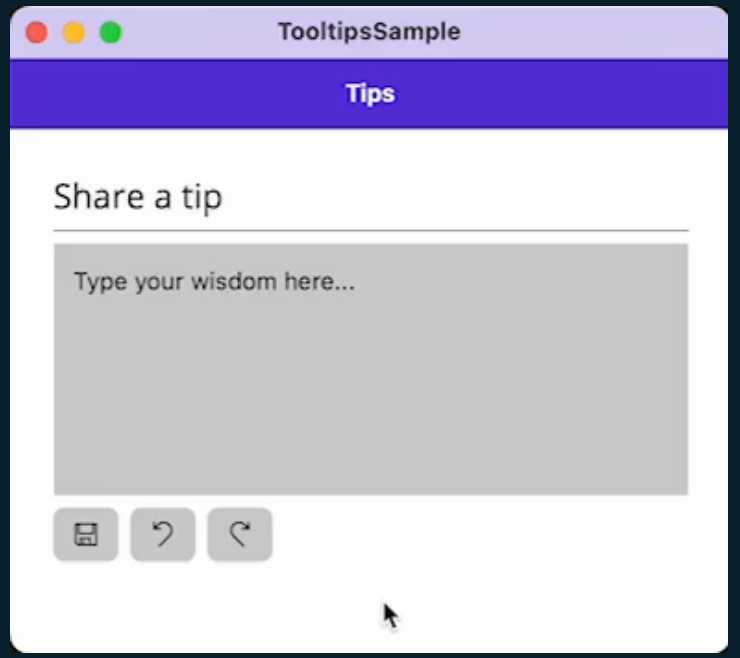
Tooltips

```
<HorizontalStackLayout Spacing="8">
    <Button Text="{x:Static icons:FabIconFont.Save}"
            ToolTipProperties.Text="Save"
            FontFamily="FabricIcon"/>
    <Button Text="{x:Static icons:FabIconFont.Undo}"
            ToolTipProperties.Text="Undo"
            FontFamily="FabricIcon"/>
    <Button Text="{x:Static icons:FabIconFont.Redo}"
            ToolTipProperties.Text="Redo"
            FontFamily="FabricIcon"/>
</HorizontalStackLayout>
```



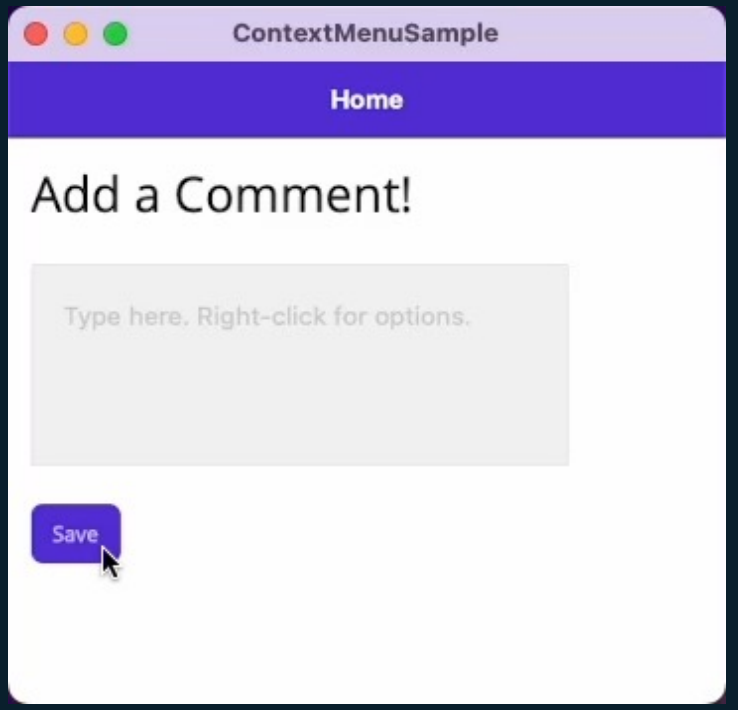
Pointer Hover

```
<Style TargetType="Button">
  <Setter Property"FontSize" Value="18"/>
  <Setter Property="VisualStateManager.VisualStateGroups">
    <VisualStateGroupList>
      <VisualStateGroup x:Name="CommonStates">
        <VisualState x:Name="Normal">
          <VisualState.Setters>
            <Setter Property="Background" Value="{StaticResource Gray200}"/>
            <Setter Property="TextColor" Value="{StaticResource Gray950}"/>
          </VisualState.Setters>
        </VisualState>
        <VisualState x:Name="PointerOver">
         <VisualState.Setters>
           <Setter Property="TextColor" Value="{StaticResource White}" />
           <Setter Property="Background" Value="{StaticResource Primary}" />
         </VisualState.Setters>
       </VisualState>
     </VisualStateGroup>
   </VisualStateGroupList>
 </Setter>
</Style>
```



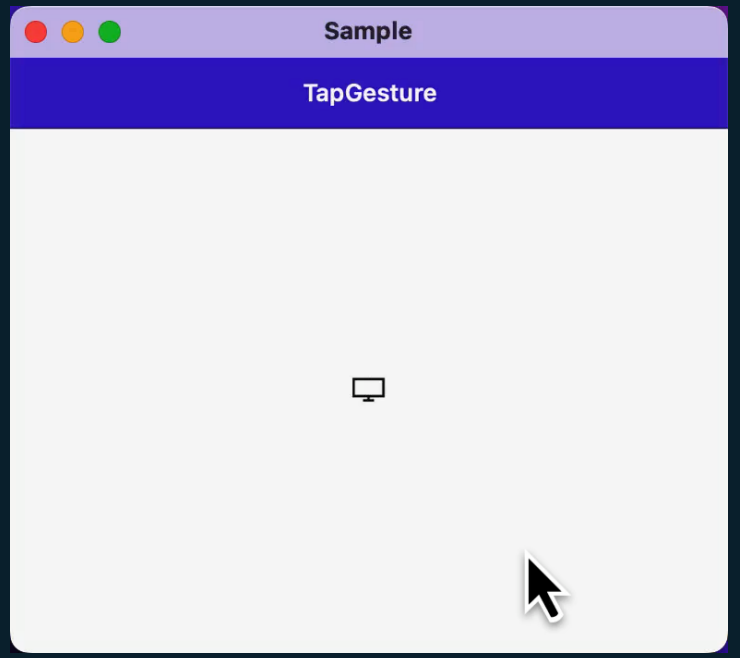
Context Menu

```
<Editor>
    <FlyoutBase.ContextFlyout>
        <MenuFlyout>
            <MenuFlyoutItem Text="Increase font"
                Clicked="OnIncrease"/>
            <MenuFlyoutItem Text="Decrease font"
                Clicked="OnDecrease"/>
            <MenuFlyoutItem Text="Clear"
                Clicked="OnClear"/>
        </MenuFlyout>
    </FlyoutBase.ContextFlyout>
</Editor>
```



Right-Click

```
<Editor.GestureRecognizers>
    <TapGestureRecognizer Buttons="Primary"
                          NumberOfTapsRequired="2"
                          Tapped="Primary_Tapped"/>
    <TapGestureRecognizer Buttons="Secondary"
                          Tapped="Secondary_Tapped"/>
</Editor.GestureRecognizers>
```



Component Vendors











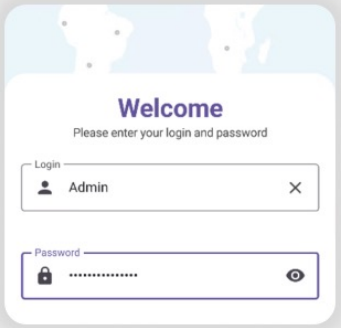


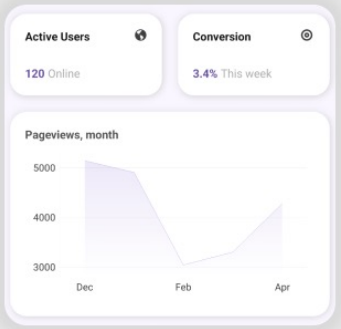




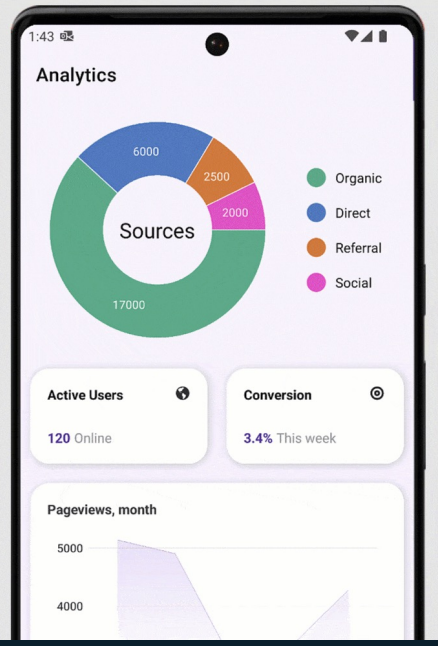


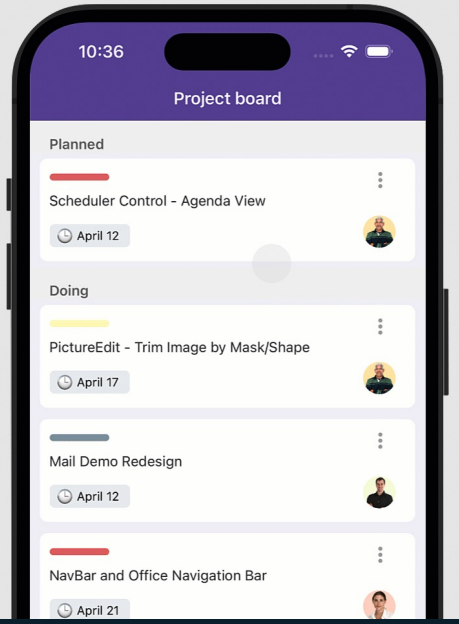
DevExpress

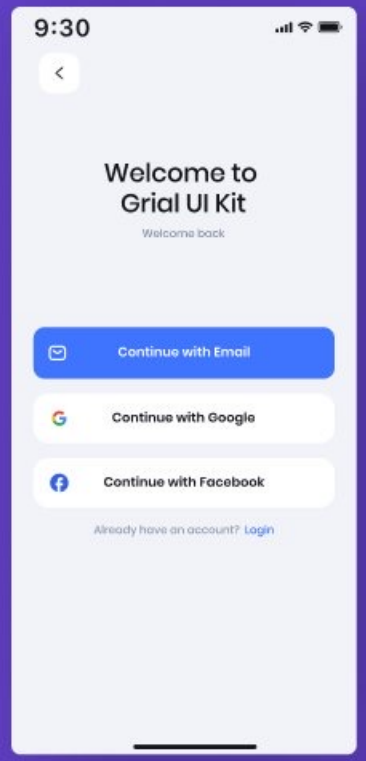


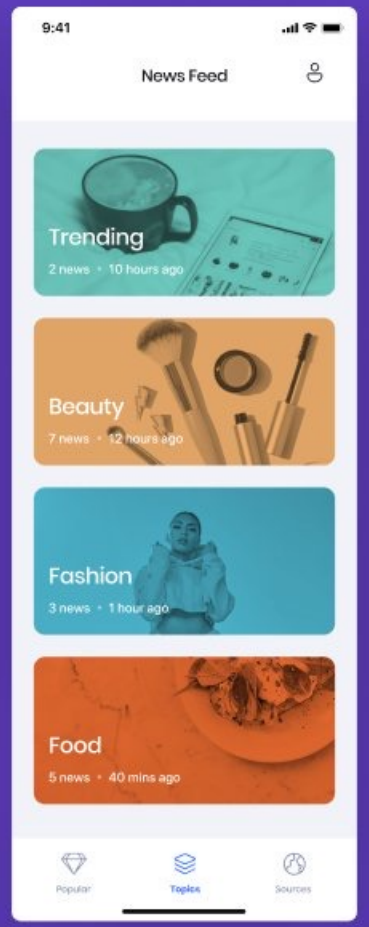


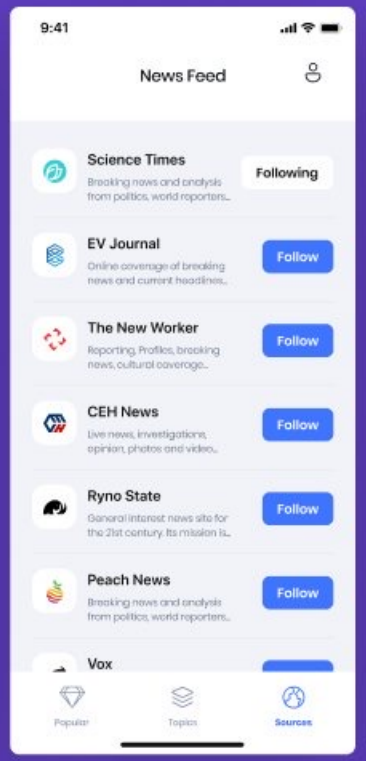
- All controls are FREE
- DataGrid, CollectionView, Filtering, CRUD
- Integration with Office File library
- Many samples and docs



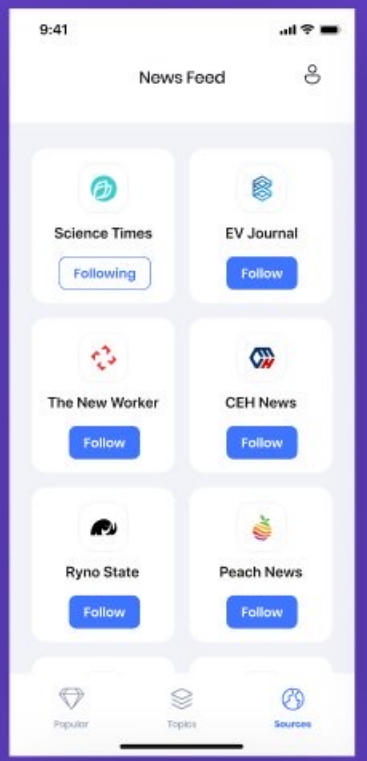


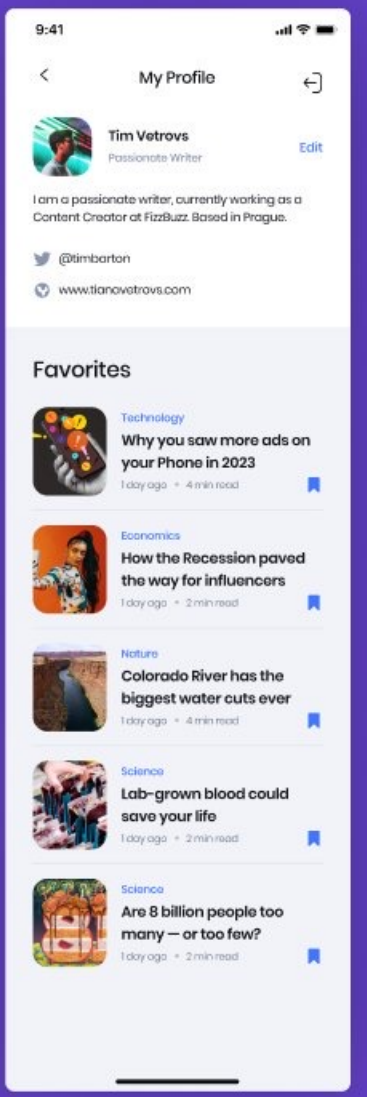

9:30
Welcome to Grial UI Kit
Welcome back
Continue with Email
Continue with Google
Continue with Facebook
Already have an account? Login

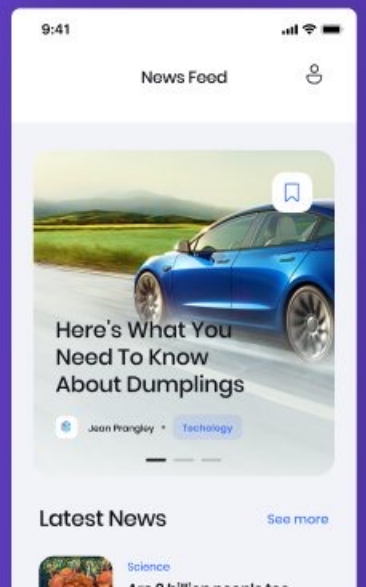

9:41
News Feed
Trending
2 news • 10 hours ago
Beauty
7 news • 12 hours ago
Fashion
3 news • 1 hour ago
Food
5 news • 40 mins ago
Popular
Topics
Sources

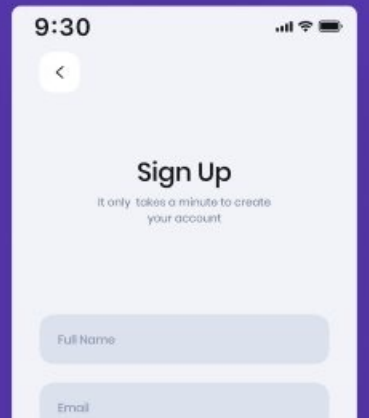

9:41
News Feed
Science Times
Following
EV Journal
Follow
The New Worker
Follow
CEH News
Follow
Ryno State
Follow
Peach News
Follow
Vox
Popular
Topics
Sources

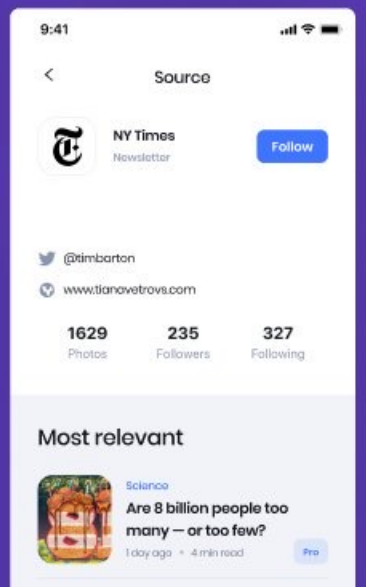

9:41
New Detail
Plug-In Car Sales Decreased 1% In March 2023 in Germany
Techology • 6 min read • Apr 19, 2023
EV Journal
Jean Prangley
Follow
In March, new passenger car registrations in Germany increased by 16.6 percent year-over-year to 281,361. During the first quarter of 2023, a total of 666,818 cars were registered (up 6.5% year-over-year).
Meanwhile, the plug-in electric car segment noted a slight decline of 1.4 percent year-over-year to 60,901 registrations. That's because of the collapse of the plug-in hybrid sales (without a $4,930/€4,500 incentive this year). Plug-in hybrids noted 18,776 registrations (down 39 percent), which is 8.0 percent of the total volume.
Well, with our homemade dumpling recipe, we're bringing that warm, fuzzy feeling right to your very own kitchen table.
INSIDEEVs
The white bathtub ring around Lake Mead
So far this year, more than 132,000 passenger plug-in electric cars were registered in Germany, which is 13 percent less than a year ago.
INSIDEEVs
The white bathtub ring around Lake Mead

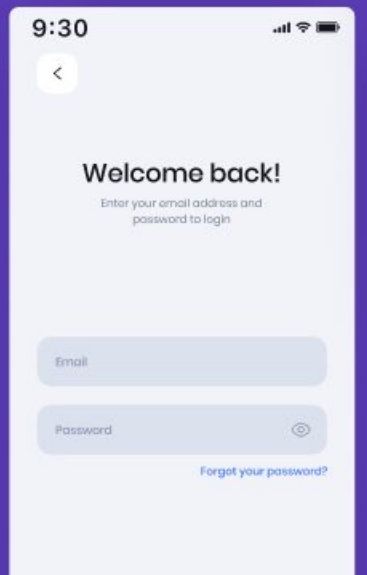

9:41
News Feed
Science Times
Following
EV Journal
Follow
The New Worker
Follow
CEH News
Follow
Ryno State
Follow
Peach News
Follow
Popular
Topics
Sources


9:41
My Profile
Tim Vetrovs
Passionate Writer
Edit
I am a passionate writer, currently working as a Content Creator at FizzBuzz. Based in Prague.
@timbarton
www.timvetrovs.com
Favorites
Technology
Why you saw more ads on your Phone in 2023
1 day ago • 4 min read
Economics
How the Recession paved the way for influencers
1 day ago • 2 min read
Nature
Colorado River has the biggest water cuts ever
1 day ago • 4 min read
Science
Lab-grown blood could save your life
1 day ago • 2 min read
Science
Are 8 billion people too many — or too few?
1 day ago • 2 min read

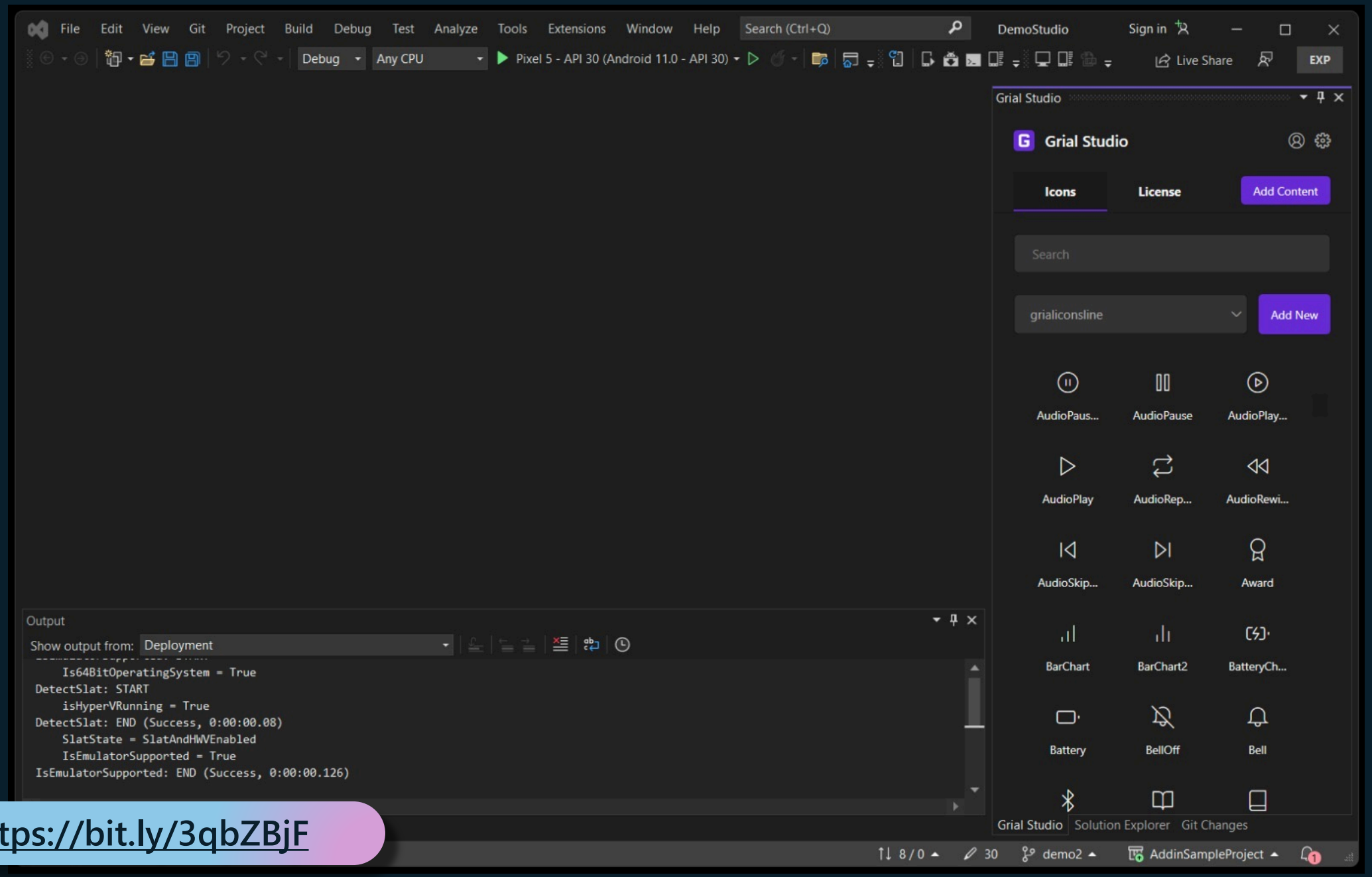

9:41
News Feed
Here's What You Need To Know About Dumplings
Jean Prangley • Techology
Latest News
See more
Science


9:30
Sign Up
It only takes a minute to create your account
Full Name
Email


9:41
Source
NY Times
Newsletter
Follow
@timbarton
www.timvetrovs.com
1629
Photos
235
Followers
327
Following
Most relevant
Science
Are 8 billion people too many — or too few?
1 day ago • 4 min read
Pro

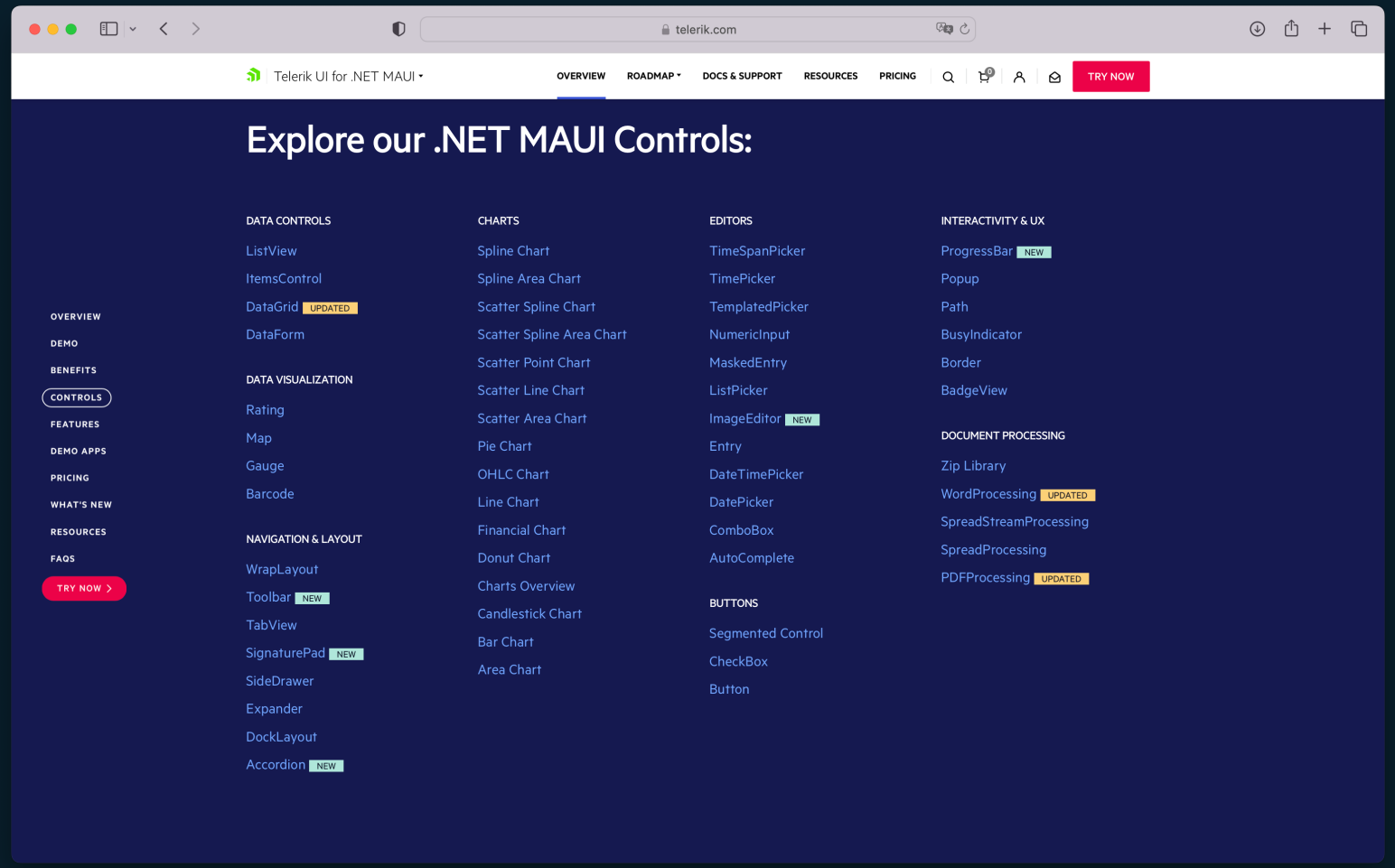

9:30
Welcome back!
Enter your email address and password to login
Email
Password
Forgot your password?

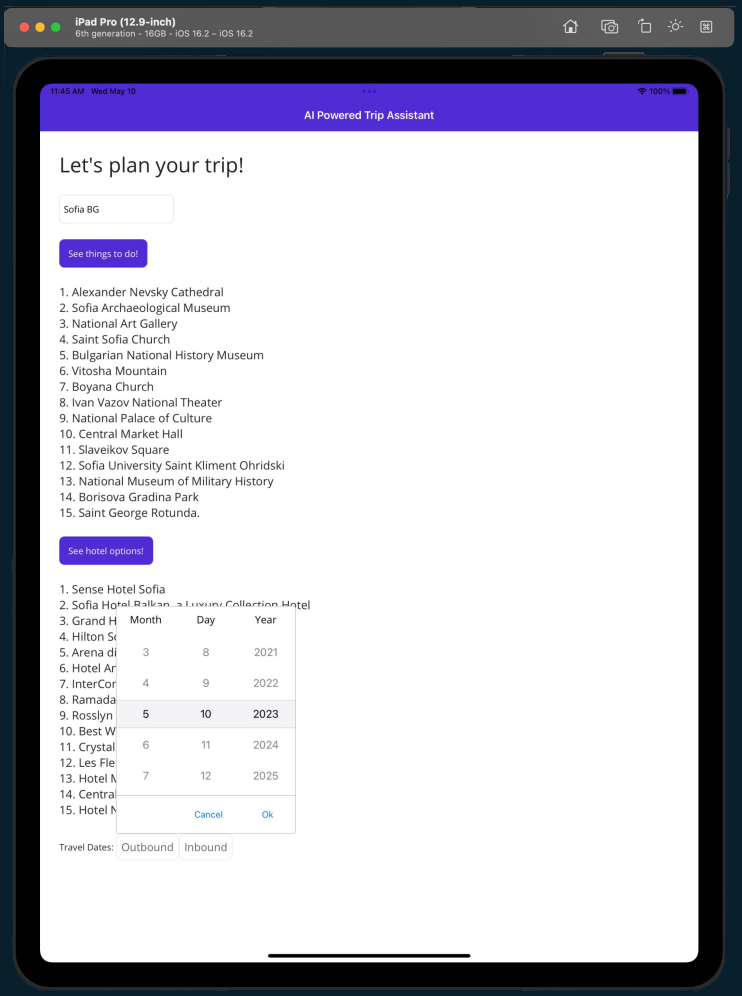
Grial



https://bit.ly/3qbZBjF

Telerik UI for .NET MAUI



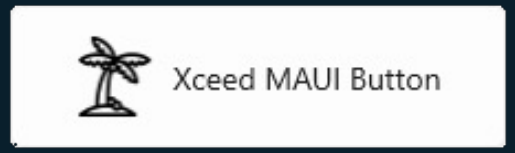


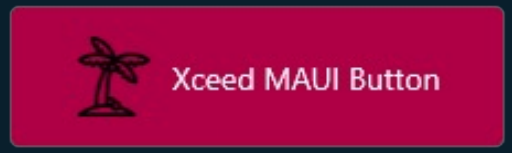
SYNCFUSION'S

.NET MAUI CONTROLS

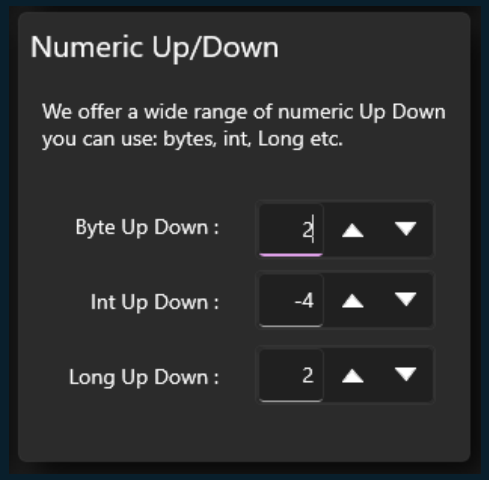


Syncfusion®

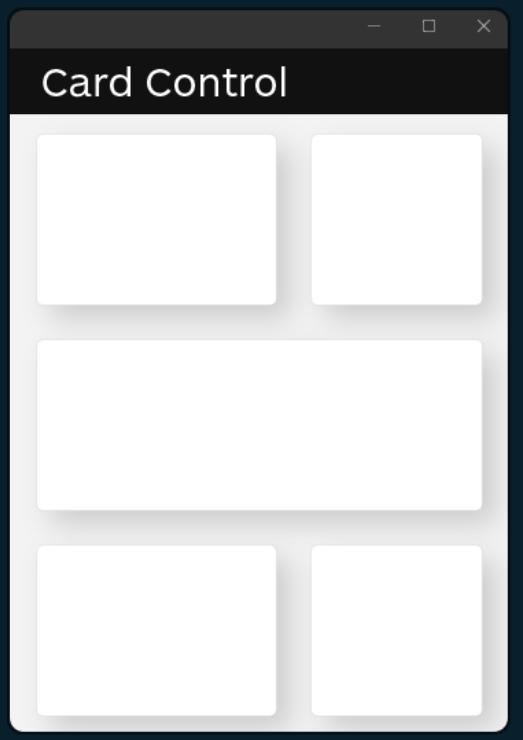












Free Controls:

- Border
- Button
- ToggleButton
- RepeatButton
- NumericUpDown
- ContentControl
- ButtonSpinner
- Card

https://github.com/xceedsoftware/mauitoolkit

Coming in .NET 8

Stability



Quality effort report for top feedback

Memory mgmt and perf improvements

Workloads



dotnet workload clean

NuGet packages for previews

Desktop



Form navigation with keyboard

Keyboard listeners

Hybrid



Deep linking for Blazor Hybrid

HybridWebView

Native

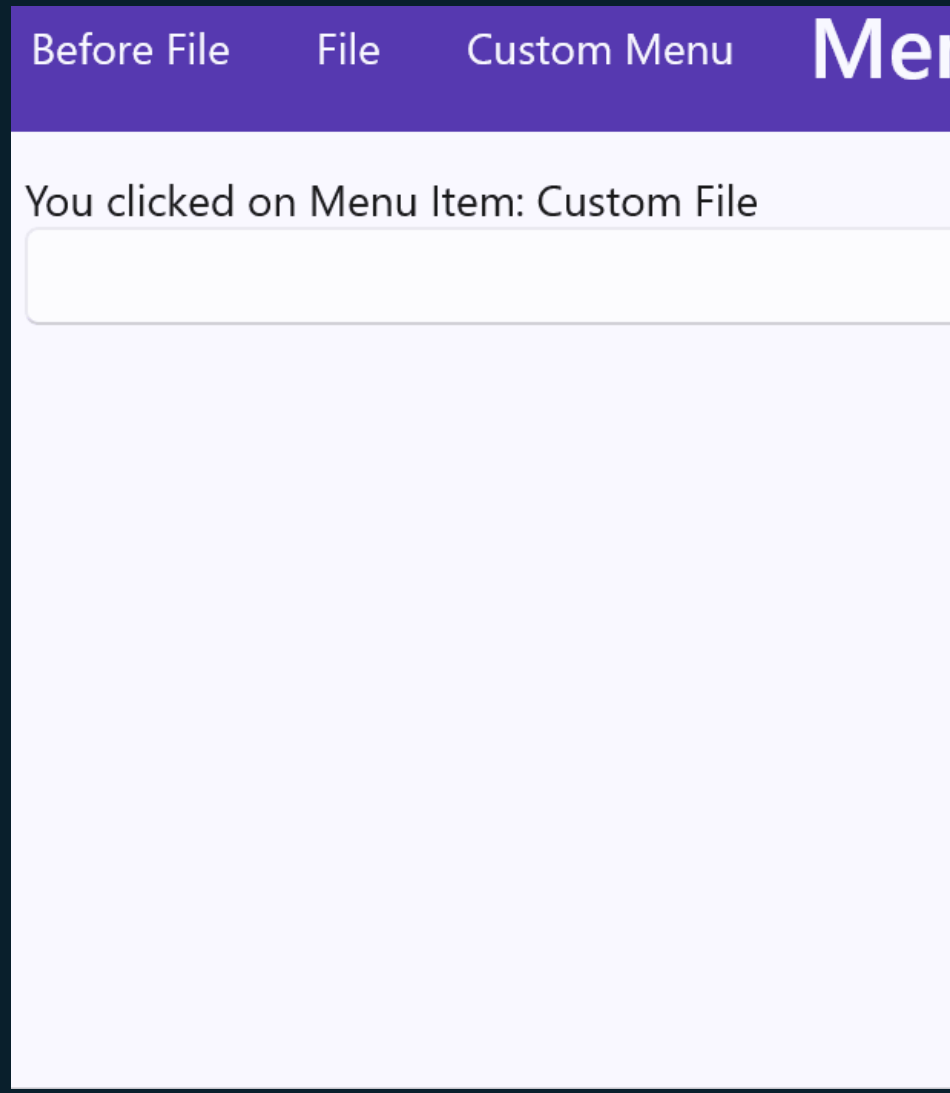


Smaller & faster apps with Native AOT

Sharing .NET libs to Obj-C and Swift

Accelerators

```
<MenuBarItem Text="File">
  <MenuFlyoutItem Text="New"
                  Accelerator="ctrl+n"/>
  <MenuFlyoutItem Text="Save"
                  Accelerator="ctrl+s"/>
  <MenuFlyoutItem Text="Save as"
                  Accelerator="ctrl+shft+s"/>
<MenuFlyoutSeparator />
  <MenuFlyoutItem Text="Exit"
                  Accelerator="ctrl+q"/>
</MenuBarItem>
```



Use Native Libraries with Platform Channels

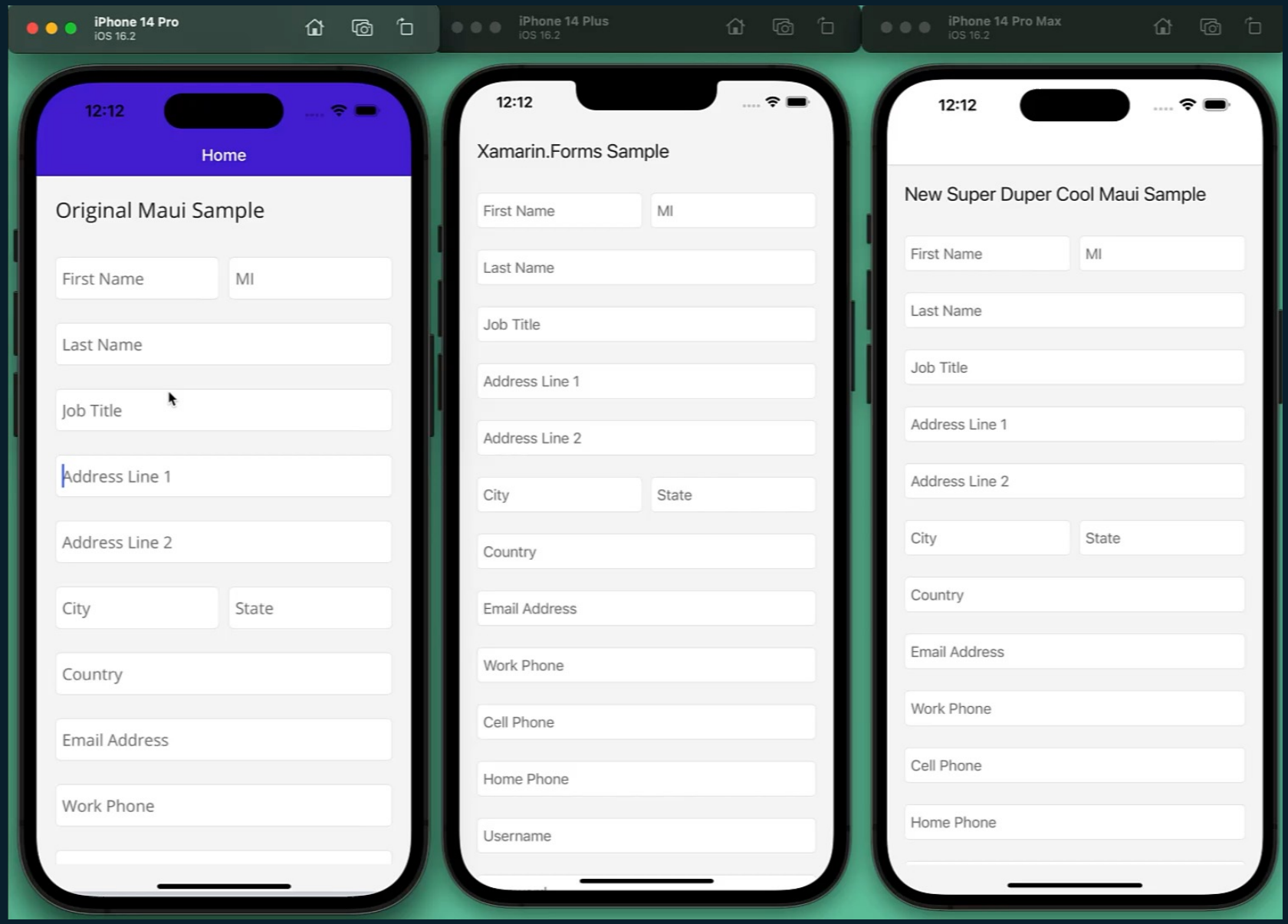
```java
public class LabelViewChannel extends ViewChannel {
  @Override
  public Object handleMessageFromDotNet(String messageId, Object... parameters) {
    if (messageId.equals("setText")){
      textView.setText((String)parameters[0]);
      return textView.getText();
    }

    return null;
  }
}
```

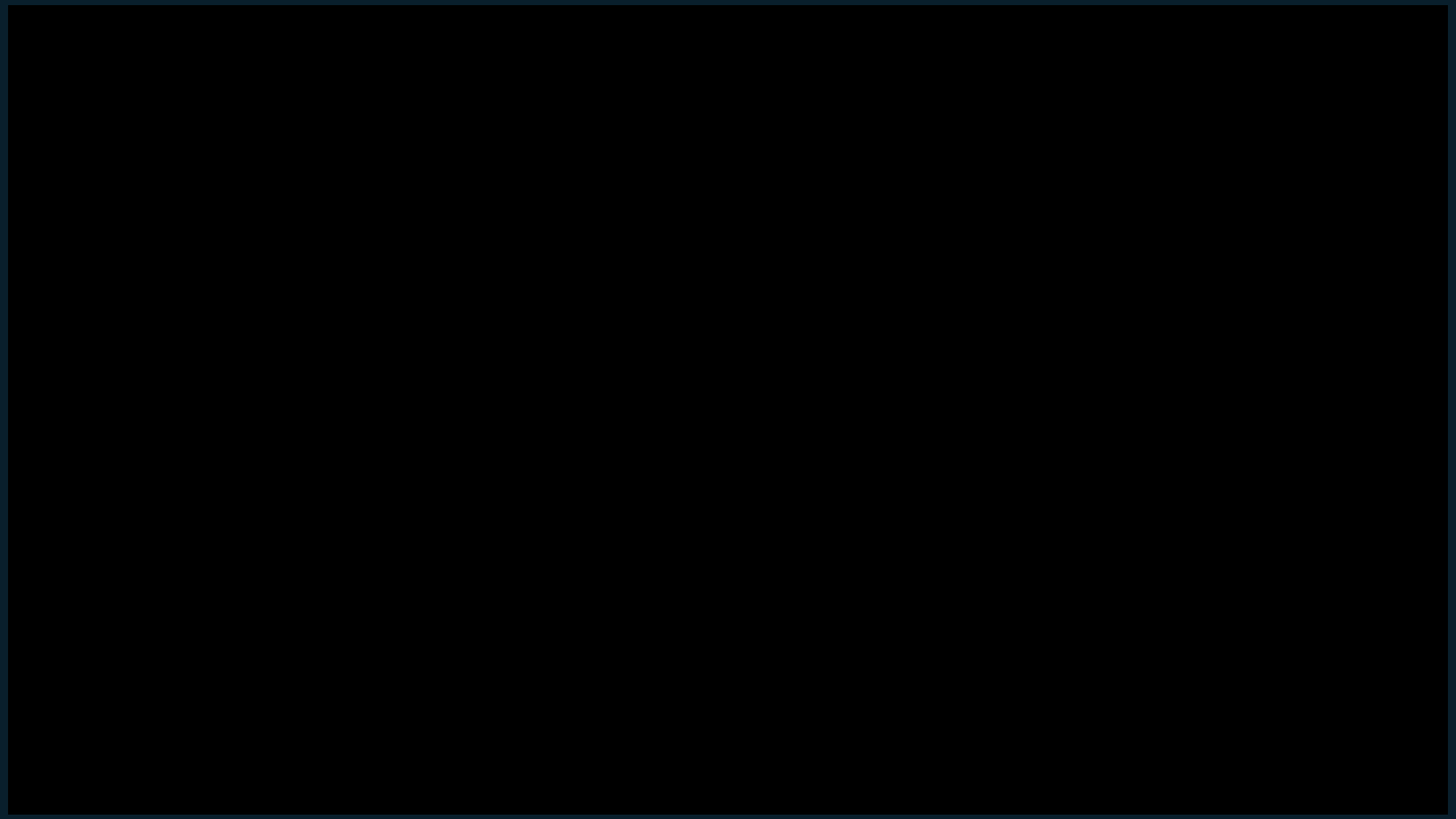
```
void Button_Clicked(System.Object sender, System.EventArgs e) {
  var labelViewChannel = ChannelService.GetOrCreateChannel("labelViewChannel");
  labelViewChannel.SendToPlatform("setText", "Hello, From MAUI!");
}
```

Keyboard Improvements

- No keyboard overlap
- Scroll to focus
- Tabbing/Next defaults



Productivity with GitHub Copilot



Experimental

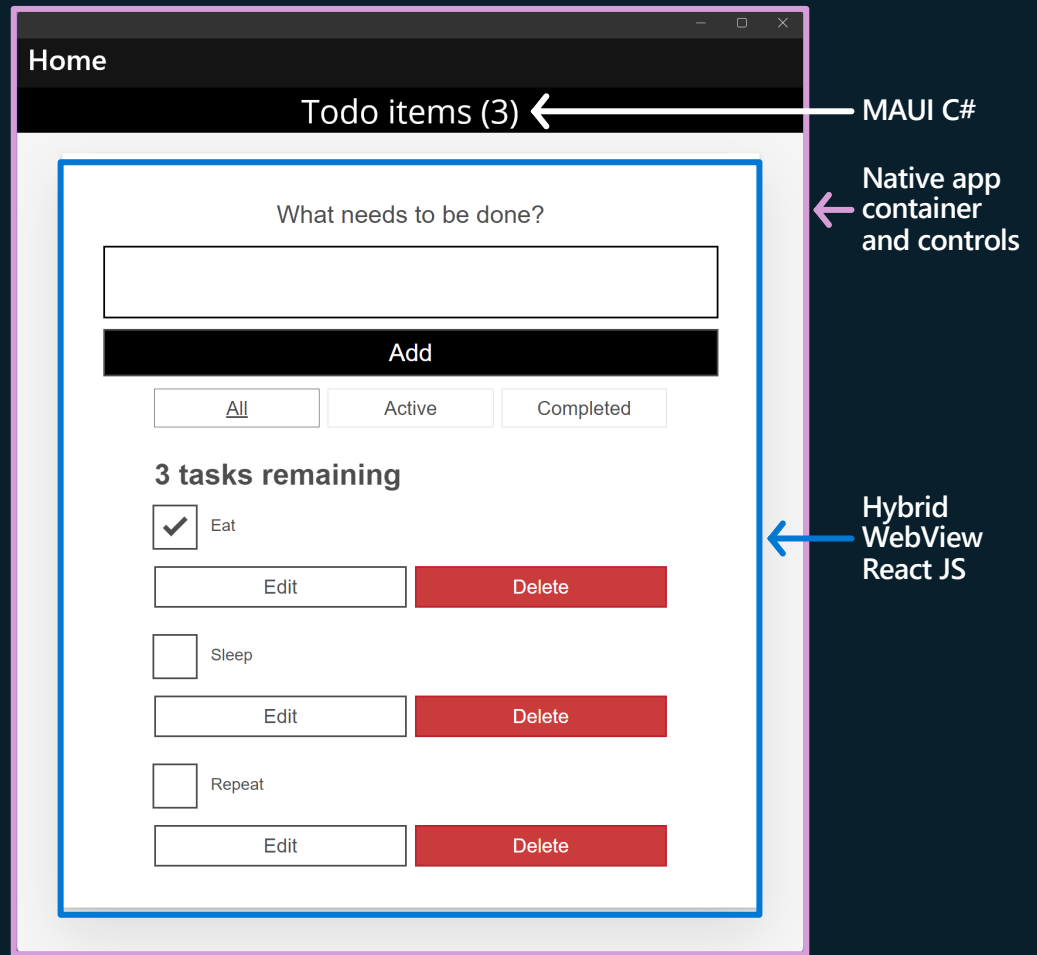
https://aka.ms/MauiHybridWebView

HybridWebView for Angular, ReactJS, Vue, etc.

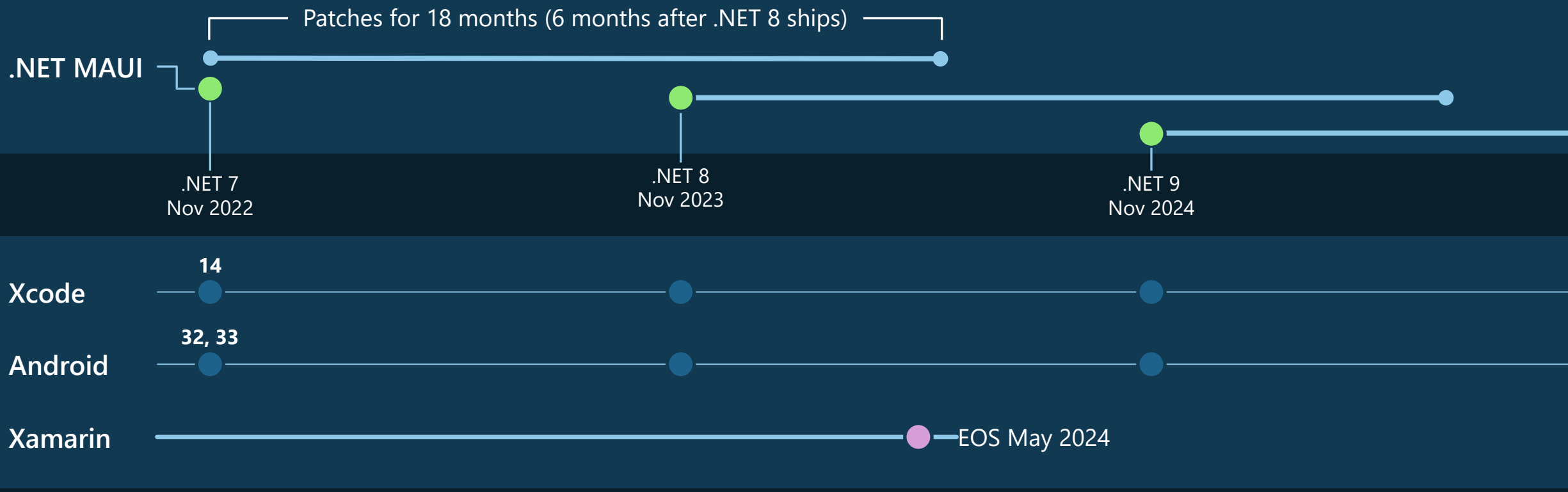
```
<Grid ColumnDefinitions="*" RowDefinitions="Auto,*">
    <Label
            Text="{Binding TodoAppTitle}"
            FontSize="32"
            HorizontalOptions="Center" />

    <hwv:HybridWebView Grid.Row="1" x:Name="myHybridWebView" HybridAssetRoot="ReactTodoApp" MainFile="index.html" />
</Grid>
```

- Multi-platform control based on .NET MAUI WebView
- Host static web assets (HTML, JS, CSS, etc.)
- Communicate between JavaScript and .NET
- Reuse your Angular, React, etc. investments in MAUI native desktop and mobile apps
- Use any web-based UI and it can interact with C# code
- Looking for feedback!



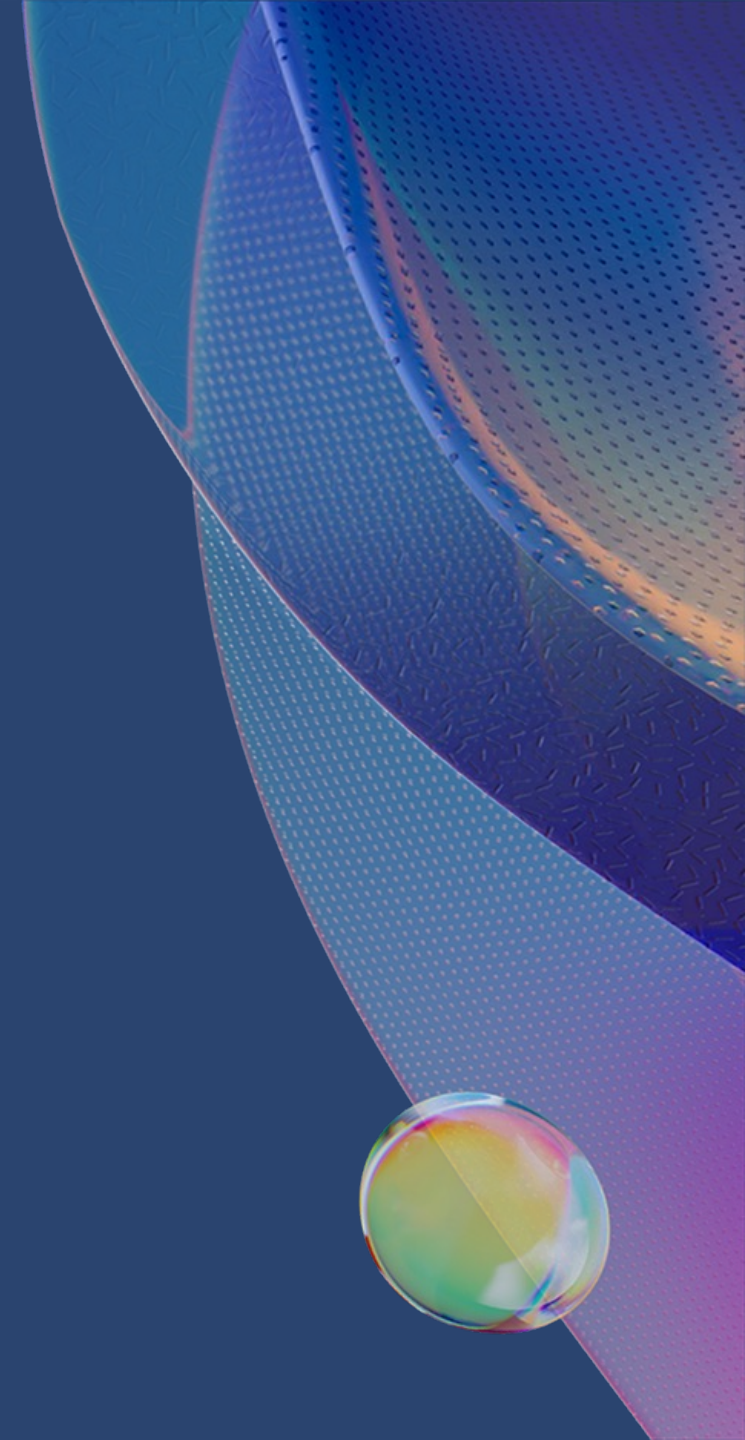
Release schedule, support & patching



- A major version of .NET MAUI will ship with .NET going forward (.NET 7, .NET 8, etc.)
- .NET MAUI is an optional workload with external dependencies (such as Xcode & Android SDK tools)
- A major version of .NET MAUI will receive patches for 6 months after a successor (major version) ships
- Xamarin SDKs will receive support until May 2024

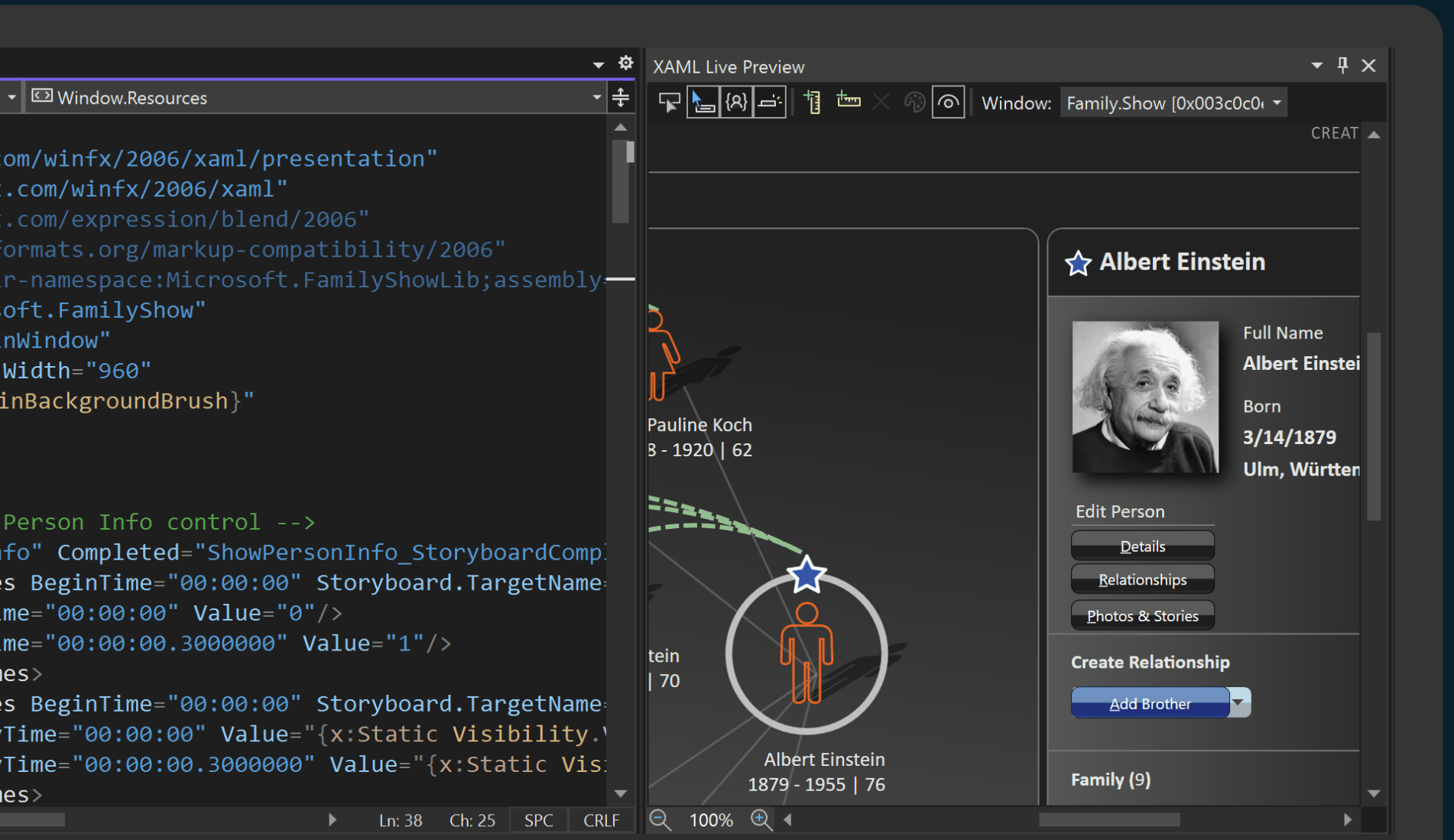
Start Using .NET MAUI

dot.net/maui



Rapid Development with Visual Studio 2022

visualstudio.microsoft.com



Rapid UI Iteration

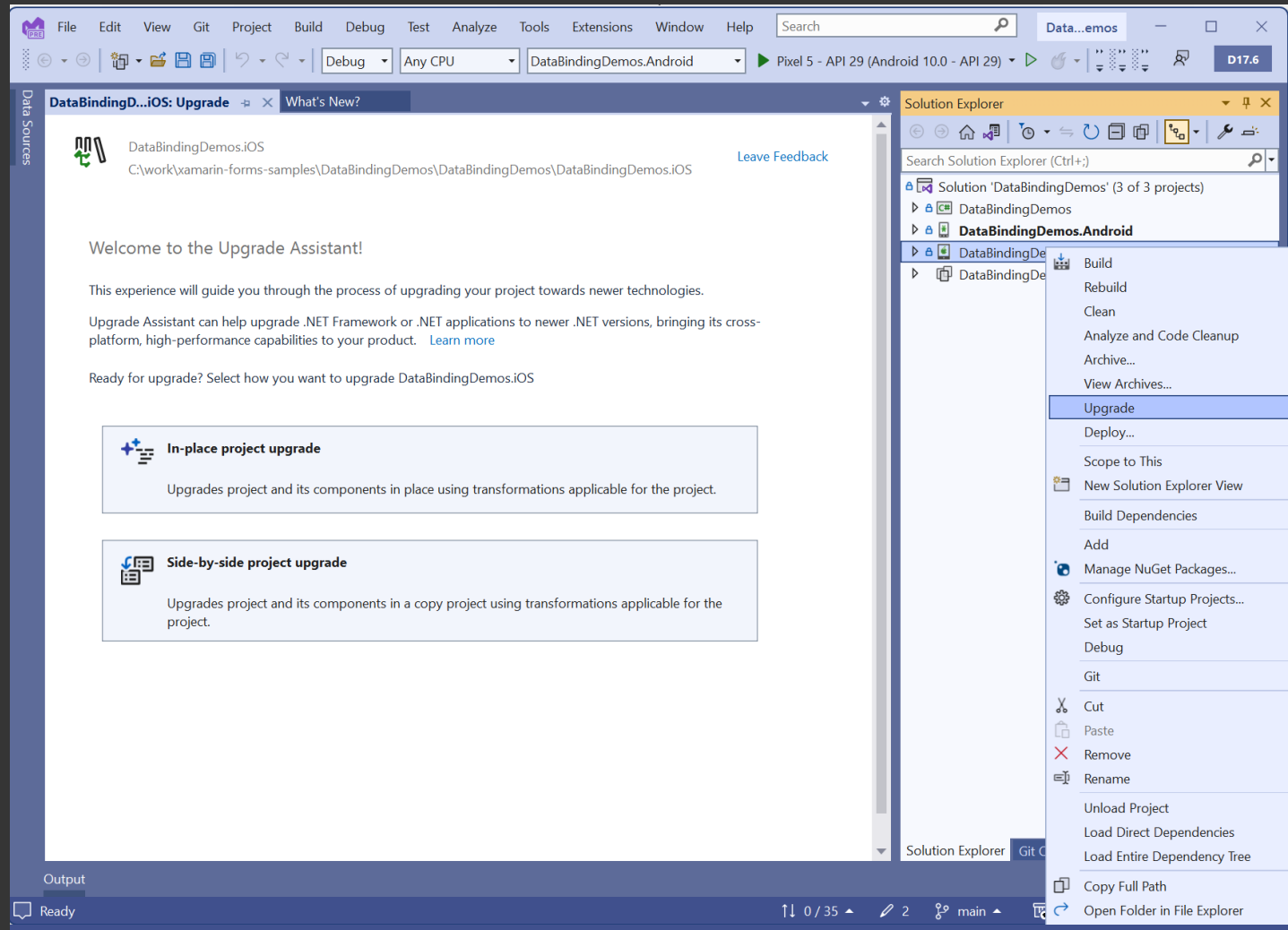
- XAML and C# Hot Reloading
- Live App Preview with UI Tree Inspection
- Remote iOS development with Pair to Mac
- Local iOS development with Hot Restart

.NET Upgrade Assistant for Xamarin Projects

Highlights:

- Converts projects
- Replaces known NuGets
- Fixes namespaces
- Fixes base types
- Applies various code updates

maui-upgrades@microsoft.com



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BRK204H: All things client and mobile app development with .NET MAUI





Microsoft

Thank You!

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